Integrating Security into Healthcare Business Processes

Challenges & Solution

VitalBridge Health Solutions

Where care meets security.

Presented by Group 5

Our team comprises dedicated professionals committed to exploring the intersection of healthcare and robust security. Through collaborative research and analysis, we aim to provide actionable insights for VitalBridge Health Solutions.

- Healthcare Security Analysts
- Process Optimization Specialists
- Compliance Experts
- Risk Management Consultants

Group Members

Collaboratively driving secure healthcare innovation.

- Eniola Ibiyemi Fatimah
- Amadi Oluchi Faith
- Naomi Oluchi Edozie
- Titilayo Kwevi
- Valentine Momanyi
- Lyne Achieno Munyolo
- Sarah Onyioze Jatto
- Ene Rita Otalu
- Adanna Somtochukwu Onu
- Dolapo Odeyemi
- Najma Abdi Omar
- Lilly Adjei
- Bilkisu Hussaini
- Javeria Hassan
- Esther Kubeh Awuku
- Emediong Bassey
- Adejuwon Ruqoyat Adebisi
- Ojimba Anita
- Priscilla Azilafu
- Deborah Orimikoya
- Joy Wekesa
- Winner Osahenrhun Ayevbuomwan
- Okafor Chinazaekpere Miracle
- Oreofe Omowunmi Dada
- Rosemary Edet
- Olasubomi Olakunle
- Abimbola Olowoporoku
- Adeoye Adedoyin
- Toyyibah Abiola Adewale

Introduction: Securing Trust in a Digital Health Ecosystem

VitalBridge Health Solutions is a rapidly expanding healthcare company specializing in telemedicine and mobile healthcare in Nigeria and globally. Our mission is to deliver **affordable, accessible, and quality care** through digital platforms, reaching underserved communities.

Digital Transformation

Improves care delivery but increases **cybersecurity & data privacy risks**.

Compliance Drivers

Nigeria Data Protection Act (NDPA 2023) & **HIPAA** are crucial frameworks.

Our Commitment

Beyond compliance, we aim to build **trust, integrity, and patient confidence**.

Core Objective

Identify challenges and foster **collaboration with CISO & Security Committee** for secure processes.

VitalBridge Health Solutions: Our Healthcare Context

As a growing healthcare provider specializing in telehealth and EHR management, VitalBridge delivers **affordable, accessible, high-quality care** digitally to thousands daily, including remote communities. We face challenges like cyber threats and data privacy, compounded by compliance needs (NDPA, HIPAA, GDPR). Protecting patient data isn't just legal; it's essential for our reputation and patient safety.

⚙️ Key Functions:

- Clinical care (Doctors, Nurses, Lab Staff)
- Remote Patient Monitoring (Wearables)
- Admin (HR, IT, Billing, Scheduling)
- Coordination with External Partners

🔄 Core Processes:

- Patient Registration & Verification
- Insurance Claims & Billing
- EHR Management
- Patient Feedback & Follow-ups

🎯 Strategic Goals:

- Build Patient Trust
- Ensure Compliance
- Deliver Quality Care
- Achieve Operational Efficiency

Why Security Must Be Embedded Early?

🌟 Reputation & Trust

Early embedding builds patient confidence; data breaches destroy it. Our commitment ensures long-term credibility.

🏥 Patient Safety

Secure data ensures accurate diagnosis and treatment. Altered records or compromised systems pose life-threatening risks.

🧩 Avoid Patchwork Risks

Integrating security from day one ensures seamless protection. Retrofitting leads to disruptions, errors, and compliance failures.

Proactive security is not a cost, but an investment in patient well-being and organizational resilience.

Key Cybersecurity Challenges in Healthcare

Challenge 1
Awareness & Training

Challenge 2
Resistance to Secuity Polices

Challenge 3
Budget Constraints

Challenge 4
Misalignment

Challenge 5
Communication Gap

Challenge 6
Legacy Systems

Challenge 7
Undefined Roles & Accountability

Challenge 8
Weak Incident Preparedness

Challenge 9
Third-Party Risks

Challenge 10
Regulatory Complexity

Navigating Security Integration: Challenges for Business Leaders

Addressing these core challenges is crucial for successful security integration without compromising patient care or efficiency.

Key Strategies for Integrating Security

🔗 Collaborative Syncs

Regular cross-functional meetings to align workflows with security needs.

🛡️ Joint Risk Assessments

Workshops to link operational risks directly with patient safety & data privacy.

📊 Shared Metrics

Combine operational & security KPIs for a holistic view of compliance and efficiency.

👥 Security Champions

Empower peer leaders within units to bridge staff and security teams effectively.

⏳ Embed Security Early

Involve CISO in new tech/projects from inception to avoid costly retrofitting.

📢 Advocate Resources

Secure essential funding for robust tools, comprehensive training, & modernizing legacy systems.

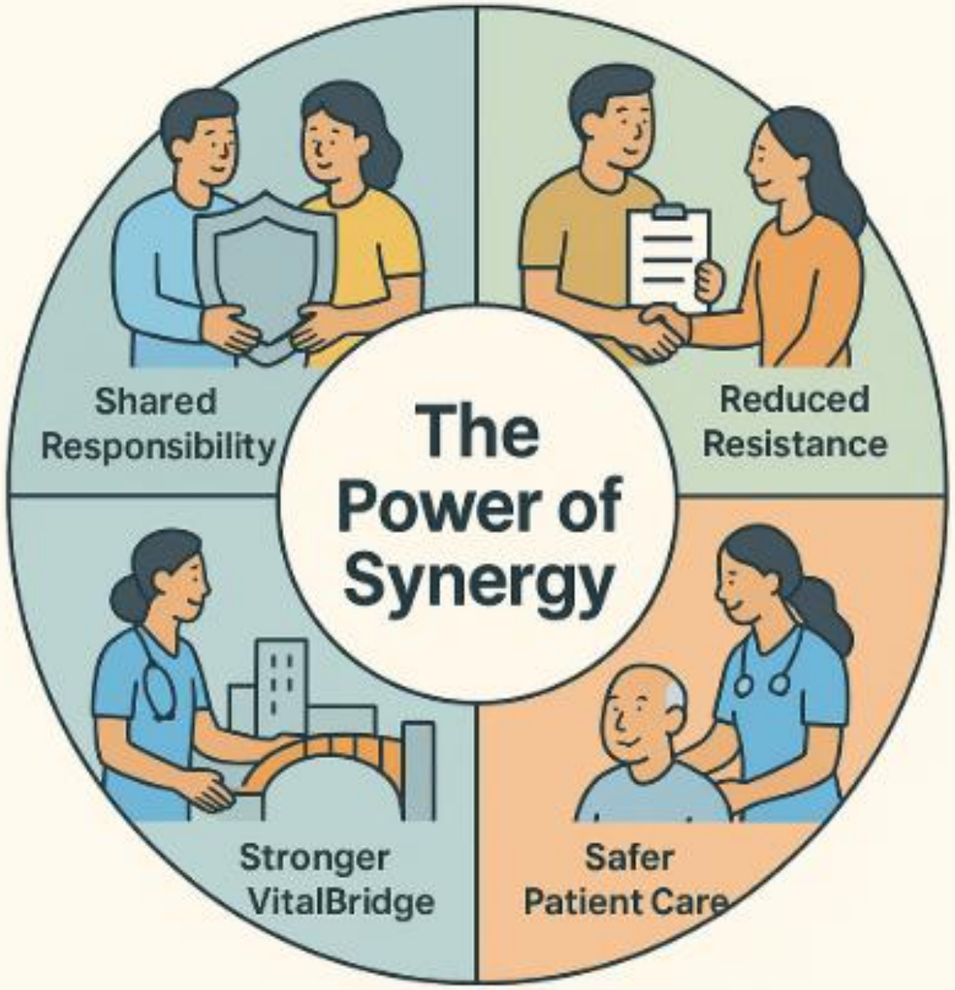
💬 Simplified Communication

Translate complex security policies into clear, actionable steps for all staff members.

Strengthening the Partnership: Business Units & Security Leadership

- **Bi-weekly Syncs:** Dedicated meetings to discuss workflow challenges and emerging risks.
- **Risk Workshops:** Collaborative sessions to prioritize risks based on their potential impact to patient care and operational continuity.
- **Shared KPIs:** Jointly developed metrics that measure both compliance and operational efficiency, ensuring alignment.
- **Security Champions:** Empowering peer-to-peer support for security adoption across all departments, fostering a culture of shared responsibility.
- **Communication Plan:** Developing clear reporting channels and easy-to-follow policies to ensure staff engagement and understanding.

The Power of Synergy: Benefits of Collaboration



Security Leadership & Collaboration at VitalBridge

🛡️ CISO (Chief Information Security Officer)

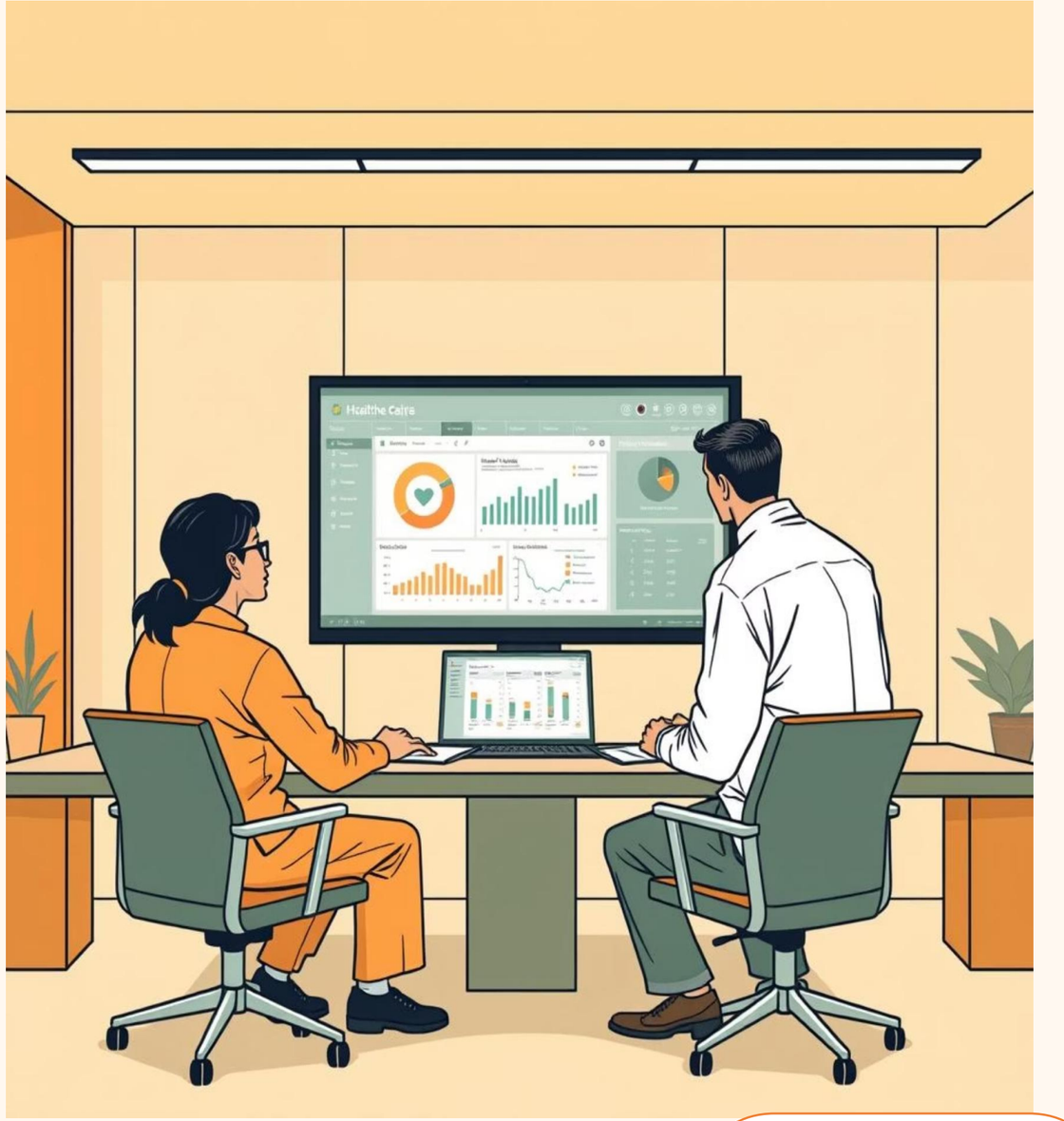
- Develops & executes InfoSec strategy (HIPAA, NDPA, GDPR).
- Leads risk management (identify, assess, prioritize threats).
- Ensures policies & controls meet standards.
- Heads incident response & crisis management.
- Ensures regulatory compliance.

🏛️ Security Committee

- Provides governance & oversight of policies.
- Prioritizes risks & projects based on business impact.
- Collective decision-making (budget, controls, incidents).
- Ensures security **enables, not blocks** business growth.

🤝 Collaboration with Business Leaders

- Embeds compliance across **clinical, IT & external partners**.
- Balances **patient care efficiency with security controls**.
- Promotes awareness & training culture.
- Aligns risk assessments with operational realities.
- Ensures proactive, coordinated incident response.



Practical & Actionable Steps

Security Education

Tailored security education for doctors, nurses & admin staff.

Periodic Reviews

Conduct reviews against HIPAA, NIST CSF, and ISO 27001 standards

Security Advocates

Appoint departmental leaders to advocate security practices

Multi-Factor Authentication

Mandate MFA for all EHR & telemedicine access

Breach Simulation

Simulate breach & social engineering scenarios to test preparedness.

Governance Roles & Alignment

🛡️ **Information Security Manager:**

- Supports InfoSec decision-making.
- Aligns IT with business goals.
- Ensures compliance & metrics.

🏛️ **Board of Directors:**

- Defines direction & approves IT initiatives.
- Final investment decisions.
- Provides accountability.

👩‍💻 **Information Security Team:**

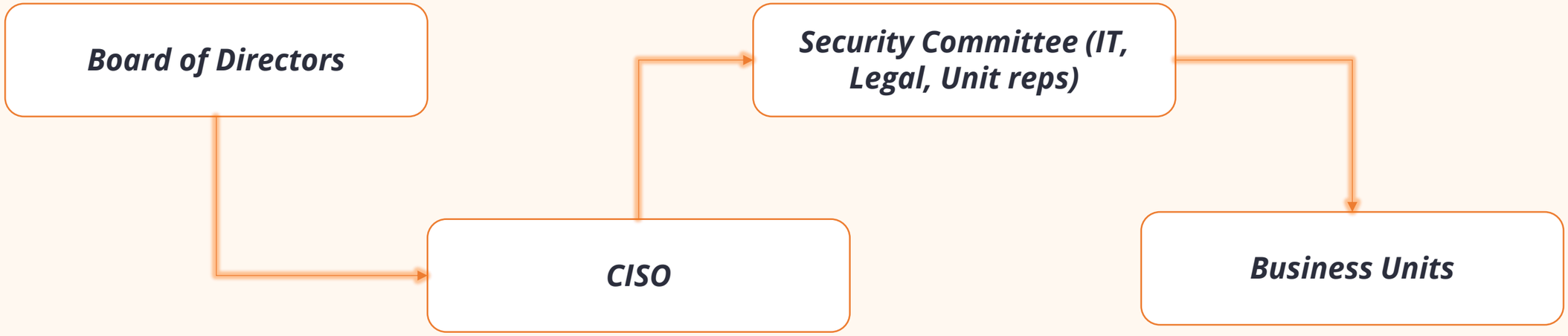
- Implements governance policies.
- Monitors IT activities & risks.
- Mitigation plans & performance tracking.

Governance Roles & Alignment (Contd.)

Importance of Governance Alignment:

Ensures accountability, strengthens daily operations, prevents silos, and links business strategy with IT and security outcomes.

Governance Structure:



Strategic Recommendations: Short-Term (0–6 Months)

01

📘 Raise Awareness & Build Knowledge

Implement mandatory training sessions and interactive workshops on critical threats like phishing, password hygiene, and secure data handling practices for all personnel.

02

📑 Align Processes with Policies

Review and adjust existing operational workflows, including patient onboarding, data access approvals, and incident reporting mechanisms, to seamlessly integrate current security standards.

03

🛡️ Appoint Security Champions

Designate and train departmental representatives to act as crucial liaisons, facilitating communication and fostering a security-first mindset between their teams and the CISO/IT security department.

04

💬 Establish Clear Communication

Develop a dedicated, accessible channel (e.g., a specific email alias or secure chat group) for rapid reporting of security incidents, direct queries, and dissemination of urgent alerts across the organization.

Strategic Recommendations: Medium-Term (6–12 Months)

🛡️ **Adopt Practical Security Tools:**

- Implement advanced access controls, conduct regular vulnerability scans, and leverage automation tools to enhance security posture.

🔄 **Integrate Security into Workflows:**

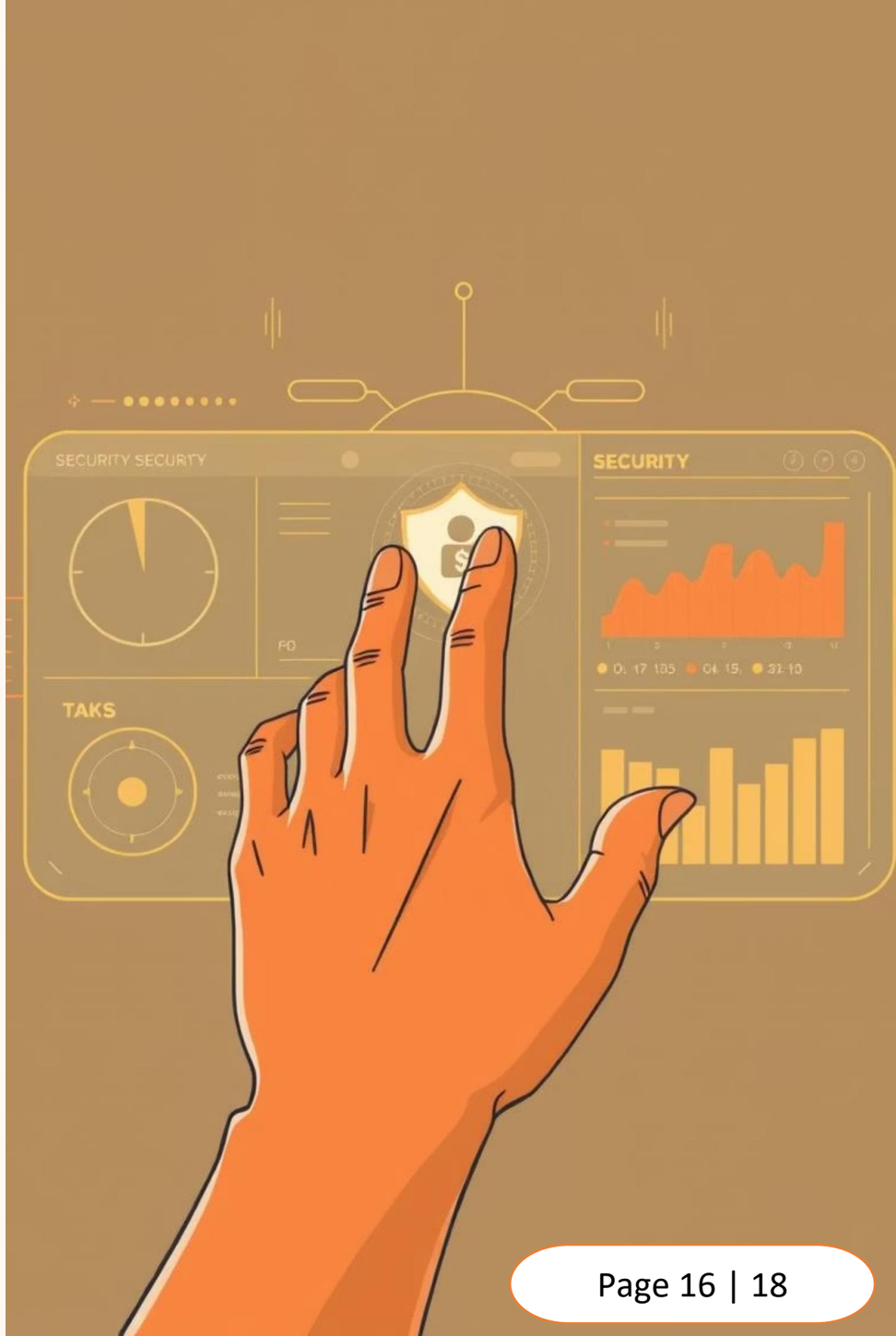
- Embed security checks into critical business processes like supplier onboarding and project approvals.

📊 **Use Dashboards & KPIs:**

- Develop interactive dashboards and Key Performance Indicators to track security incidents, training completion rates, and compliance scores in real-time.

🤝 **Strengthen Collaboration:**

- Organize quarterly review meetings with the CISO and the Security Committee to assess progress and address new challenges.



Strategic Recommendations: Long-Term (12–24+ Months)

1

🛡️ Embed “Security by Design”

Integrate security considerations from the initial stages of all new products, services, and projects, ensuring inherent protection.

2

👥 Foster a Security-Conscious Culture

Leadership actively reinforces and rewards best practices, creating an environment where security is a core value.

3

🏛️ Integrate with Governance Frameworks

Align security practices with established frameworks such as HIPAA, ISO 27001, NIST, and COBIT for comprehensive compliance.

4

⚙️ Invest in Automation & Innovation

Implement AI analytics, real-time monitoring solutions, and automated compliance checks to enhance efficiency and threat detection.

Conclusion: Secure Growth for VitalBridge

Integrating **security into business operations** is not just a necessity but a strategic imperative for VitalBridge Health Solutions' sustainable growth and long-term resilience.

> Collaboration between the **CISO, Security Committee, and business leaders** ensures robust governance, proactive oversight, and effective risk management across all levels of the organization.

Security is redefined not merely as a **technical safeguard**, but as a fundamental **enabler of trust, reputation, and innovation** in the healthcare landscape.

By consistently following the **practical, actionable steps** recommended, VitalBridge can significantly strengthen its compliance posture, rigorously protect sensitive patient data, and ultimately uphold its mission of delivering secure, high-quality healthcare.